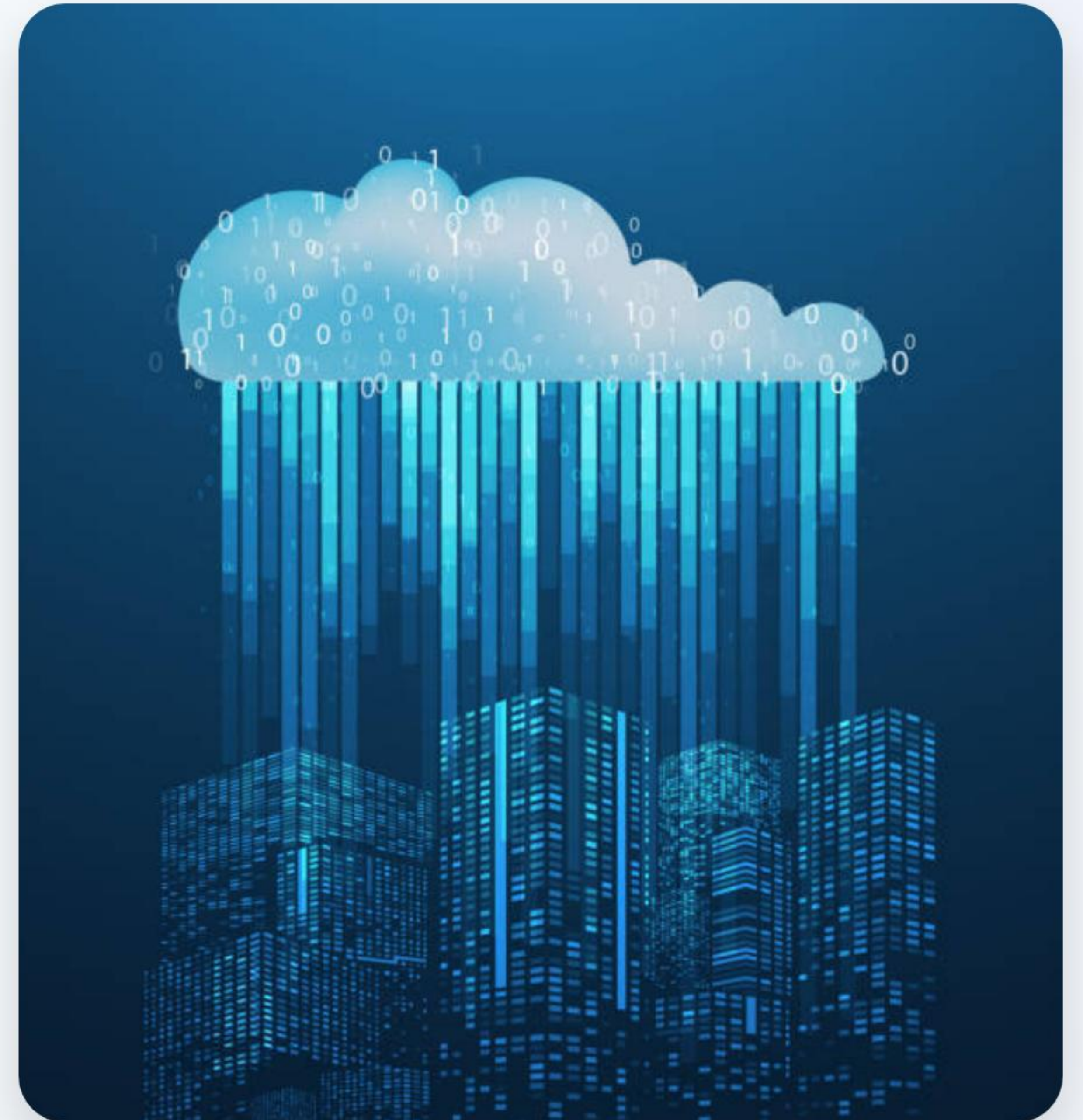


 CONCEPT ASSESSMENT

Cloud Computing core architecture & metrics

Test your knowledge on paradigms, operational indicators, isolation characteristics, billing profiles, and architecture components.



Which represents the correct hierarchical order of cloud building blocks from the top layer (closest to the end-user) to the bottom layer (closest to the provider's physical hardware)?

- A** Application software, Development platforms, Resource sharing, Infrastructure
- B** Infrastructure, Resource sharing, Development platforms, Application software
- C** Application software, Resource sharing, Development platforms, Infrastructure
- D** Development platforms, Application software, Resource sharing, Infrastructure

Which represents the correct hierarchical order of cloud building blocks?

A Application software, Development platforms, Resource sharing, Infrastructure

📄 1. APPLICATION SOFTWARE (END-USER)

>_ 2. DEVELOPMENT PLATFORMS

📦 3. RESOURCE SHARING

🖥️ 4. INFRASTRUCTURE (PHYSICAL HARDWARE)

i Explanation

This correctly outlines the cloud stack where application software sits at the top for end-users, followed by developer platforms, resource sharing mechanisms, and underlying physical infrastructure.

What primary mechanism allows multiple hardware images (like VMs) and system images (like general-purpose OSs) to run securely side-by-side on a single physical infrastructure block?

- ☒ A Vendor lock-in APIs
- ☐ B Thin-client load balancing
- ☐ C Prorated billing protocols
- ☐ D Resource sharing isolation properties

What mechanism allows secure concurrent side-by-side execution of VMs?

D Resource sharing isolation properties

A Vendor lock-in APIs

B Thin-client load balancing

C Prorated billing protocols

i Explanation

Isolation properties provided by a combination of hardware and software techniques ensure security, resource, and failure boundaries between concurrent images.

An enterprise wants to deploy an application but must guarantee that they can completely customize the underlying hardware and operating system performance. According to the cloud service paradigms, which option should they select?

A Database as a Service (DBaaS)

B Platform as a Service (PaaS)

C Development Platform APIs

D Infrastructure as a Service (IaaS)

Which model permits total underlying custom hardware/OS configuration?

D Infrastructure as a Service (IaaS)

A Database as a Service (DBaaS)

B Platform as a Service (PaaS)

C Development Platform APIs

i Explanation

IaaS provides configurable CPU, memory, and disk instances, offering the most flexibility to install and configure any platform or software.

A single version of an application with a single configuration running at the provider's side to serve multiple distinct customers describes which architecture?

- A** Non-prorated capacity model
- B** Multitenant architecture
- C** Distributed IaaS instance layout
- D** PaaS API virtualization

A single app instance and configuration serving multiple distinct customers is:

B Multitenant architecture

A Non-prorated capacity model

C Distributed IaaS instance layout

D PaaS API virtualization

i Explanation

Multitenant architecture uses a single version and configuration of an application to serve multiple customers, known as tenants.

Which scenario highlights a situation where Software as a Service (SaaS) is NOT a recommended choice for an organization?

- A When there is a massive short-term need for collaboration
- B When applications demand a high level of offline access to data
- C When the application is highly standardized, such as corporate email
- D When user access spikes significantly only once a month

In which scenario is SaaS not a recommended choice?

B When applications demand a high level of offline access to data

A Short term needs

C Standardized email

D Spiky billing access

i Explanation

SaaS applications typically depend on internet/web connectivity, making them poor fits if continuous offline access is needed.

How does Platform as a Service (PaaS) primarily differ from Software as a Service (SaaS) in terms of its target audience and delivery?

- ☐ A PaaS provides physical servers rather than web access
- ☐ B PaaS requires local software installations on developer workstations
- ☐ C PaaS delivers an environment and tools for creating software rather than a completed app
- ☐ D PaaS avoids utilizing multitenant architectures entirely

How does PaaS mainly differentiate from SaaS?

- C** PaaS delivers tools and environments to build software rather than a ready-to-run app

i Explanation

PaaS delivers development platforms, APIs, and tools over the web so developers can build and host software, whereas SaaS delivers the finished application.

An enterprise spins up 100 virtual machine instances for exactly 15 minutes. If their cloud provider bills on a non-prorated basis, how will this usage be charged?

- ☐ A Charged for raw storage capacity usage only
- ☐ B Charged for 1 full hour for each of the 100 instances
- ☐ C Billed for a fraction (0.25) of an hour per instance
- ☐ D Pricing will be calculated based on function executions

How does non-prorated billing treat a 15-minute usage window?

B The enterprise will be charged for 1 full hour for each of the 100 instances

C Billed for a fraction (0.25) of an hour

i Explanation

On a non-prorated basis, each partial hour utilized is billed as a full hour, making a 15-minute window count as a whole hour.

Which of the following describes a prominent risk associated with Platform as a Service (PaaS) that might affect long-term software portability?

- A** The burden of patching database hardware manually
- B** High upfront capital expenditure licensing costs
- C** Inability to use modern visual developer layout tools
- D** Vendor lock-in due to varying proprietary APIs across providers

What key risk impacts long-term software portability in PaaS?

D

Vendor lock-in due to varying proprietary APIs across providers

i Explanation

Because PaaS APIs can vary drastically from one provider to another, using proprietary features can cause trouble if an app needs migration later.

In the 'Shared Responsibility' model of Infrastructure as a Service (IaaS), which of the following is typically managed by the cloud provider rather than the user?

- A** The configuration of user access permissions within the guest OS
- B** Physical maintenance and rack-mounting of compute, storage, and networking hardware
- C** The installation of custom runtime libraries and application patches
- D** The validation routines and application endpoints

What falls under the provider's scope in an IaaS shared model?

- B** Physical maintenance and rack-mounting of hardware

i Explanation

Physical infrastructure is deployed on the cloud provider's side, and they manage the physical hardware blocks.

According to cloud lecture standards, how is application storage space at the infrastructure layer typically measured and billed to clients?

- ☐ A Based on the number of concurrent end-users accessing files
- ☐ B By the total number of execution requests hitting an API
- ☐ C Via a standardized upfront software licensing flat fee
- ☐ D In terms of capacity usage, such as per gigabyte or terabyte

How is cloud storage infrastructure capacity typically billed?

D In terms of capacity usage, such as per gigabyte or terabyte

i Explanation

The text explicitly states that application storage is typically charged based on capacity metrics like gigabytes or terabytes.

A key business driver for adopting Software as a Service (SaaS) instead of traditional software models is the reduction of 'CapEx' (Capital Expenditure). How does SaaS achieve this?

- ☒ A By replacing upfront software license purchases with ongoing subscription fees
- ☐ B By removing the necessity of network communication lines
- ☐ C By requiring clients to maintain their own backup server clusters on site
- ☐ D By restricting access to a single user workstation in an office

How does SaaS reduce Capital Expenditure (CapEx) metrics?

- A By replacing upfront software license purchases with ongoing subscription fees

i Explanation

SaaS reduces capital expenditures by eliminating massive upfront costs and utilizing operational subscription billing.

Why do major public cloud providers establish dozens of distinct geographical regions across multiple countries instead of keeping a single massive datacenter?

- A** To prevent developers from running into vendor lock-in traps
- B** To accommodate varying local laws, regulatory compliance, and performance latency needs
- C** To bypass the need for any resource isolation properties
- D** Because server virtualization only functions when machines are physically close

Why design a geographically distributed infrastructure network?

- B** To accommodate varying local laws, regulatory compliance, and performance latency needs

Explanation

Geographical diversity helps manage localized regulations (e.g., stopping offshore data flow) and improves user experience by lowering network latency.

What core characteristic of Azure Functions' pricing model reflects standard Platform as a Service (PaaS) monetization schemes?

- A** Billing clients based on the number of storage disks physically mounted
- B** Charging based on total number of executions and execution time duration
- C** Charging purely for the physical rack dimensions leased out
- D** Enforcing a rigid upfront licensing contract lasting several years

How is serverless billing structured in PaaS environments?

- B** Charging based on total number of executions and execution time duration

i Explanation

PaaS pricing schemes often align directly with platform utilization metrics, such as execution counts and runtime totals.

The famous phrase "There is no cloud, it's just someone else's computer" points conceptually to which structural truth about cloud infrastructure?

- A All data on a cloud network is fully public and accessible by anyone
- B Cloud providers do not utilize automated platform upgrade suites
- C Every virtual service ultimately relies on physical compute, storage, and network assets
- D Using cloud solutions removes the need for application programming interfaces

What does "someone else's computer" emphasize?

- C** Every virtual service ultimately relies on physical compute, storage, and network assets owned by a provider

i Explanation

Despite the high levels of abstraction, all cloud solutions run on real physical server farms deployed on the provider's side.

A startup company expects volatile, unpredictable traffic patterns due to seasonal marketing spikes. Which cloud characteristic serves this requirement best?

A Agility and elasticity for dynamic resource scaling

B Proprietary offline data engines

C Vendor lock-in constraints

D Non-prorated architectural footprints

Which core characteristic addresses unpredictable, spiked user traffic?

A Agility and elasticity for dynamic resource scaling

i Explanation

Elasticity allows resource environments to rapidly grow or shrink to match shifting user traffic spikes on-demand.

Assessment Complete

Knowledge Assessment Review

A comprehensive review of Cloud Infrastructure, Platform paradigms (PaaS), Software distribution paradigms (SaaS), Billing Models, and Scalability Mechanisms.

 Press space or arrow keys to navigate the concept deck.

IMAGE SOURCES



<https://media.istockphoto.com/id/1256275100/vector/abstract-futuristic-smart-city.jpg?s=612x612&w=0&k=20&c=fNjwsoQ-BXNrHHs8WDIs3hFE0D2Y0SkyeqDz2KYhXFY=>

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